Physics OMR Test - Complete Question Paper with Solutions

I'll create a comprehensive PDF-ready document with all 50 questions across all 5 sections.

📋 PHYSICS OMR TEST

Date: February 11, 2026, 11:34 AM (IST)
Duration: 2 Hours (120 Minutes)
Total Questions: 50
Total Marks: 200
Maximum Attempts: 100
Timezone: Asia/Calcutta

📌 GENERAL INSTRUCTIONS

1. This paper consists of **5 Sections (A to E)** with **10 questions each**.
2. All questions carry **+4 marks** for correct answer. **No negative marking.**
3. Use **blue/black pen** only for OMR sheet.
4. Do not use whitener on the OMR sheet.
5. Rough work should be done on the sheets provided.

Section	Type	Questions	Marks per Q	Total Marks
A	Single Correct MCQ	Q1 – Q10	+4 / -0.00	40
B	Multiple Correct MCQ	Q11 – Q20	+4 / -0.00	40
C	Numerical Value	Q21 – Q30	+4 / -0.00	40
D	Subjective	Q31 – Q40	+4 / -0.00	40
E	True / False	Q41 – Q50	+4 / -0.00	40

SECTION A: SINGLE CORRECT MCQ (Q1 – Q10)

Instructions: Each question has four options. Only **ONE** option is correct. Mark the correct option on the OMR sheet.

Q1. A body is thrown vertically upward with a velocity of 20 m/s. The maximum height reached by the body is ($g = 10\ m/s^2$):

(A) 10 m
(B) 20 m ✅
(C) 30 m
(D) 40 m

Answer: **(B) 20 m**

Solution:
Using the equation: $v^2 = u^2 - 2gh$

At maximum height, $v = 0$

$0 = (20)^2 - 2(10)(h)$

$0 = 400 - 20h$

$h = 400/20 =$ **20 m**

Q2. The SI unit of Planck's constant is:

(A) J·s ✅
(B) J/s
(C) J·m
(D) J/m

Answer: **(A) J·s**

Solution:
Planck's constant h relates energy and frequency: $E = h\nu$

$h = E/\nu$

Units: h = Joule / (1/second) = Joule × second = **J·s**

The value of $h = 6.626 \times 10^{-34}$ J·s

Q3. A car moves with uniform velocity of 36 km/h for 5 seconds. The distance covered is:

(A) 50 m ✅
(B) 100 m

(C) 180 m
(D) 36 m

Answer: **(A) 50 m**

Solution:
First convert velocity: 36 km/h = 36 × (5/18) = 10 m/s

Distance = velocity × time = 10 × 5 = **50 m**

Q4. The dimensional formula of force is:

(A) $[MLT^{-1}]$
(B) $[MLT^{-2}]$ ✅
(C) $[ML^2T^{-2}]$
(D) $[ML^{-1}T^{-2}]$

Answer: **(B) $[MLT^{-2}]$**

Solution:
Force = mass × acceleration

F = m × a = m × (velocity/time) = m × (displacement/time²)

Dimensional formula = $[M] \times [LT^{-2}]$ = **$[MLT^{-2}]$**

Q5. Two bodies of masses 1 kg and 4 kg have equal kinetic energy. The ratio of their momenta is:

(A) 1:4
(B) 1:2 ✅
(C) 4:1
(D) 2:1

Answer: **(B) 1:2**

Solution:
Kinetic energy: $KE = p^2/(2m)$, so $p = \sqrt{(2mKE)}$

Since $KE_1 = KE_2 = KE$,

$p_1/p_2 = \sqrt{(2m_1KE)}/\sqrt{(2m_2KE)} = \sqrt{(m_1/m_2)} = \sqrt{(1/4)}$ = **1/2**

Ratio = **1:2**

Q6. The escape velocity from the surface of the Earth is approximately:

(A) 7.9 km/s
(B) 11.2 km/s ✅
(C) 15.0 km/s
(D) 3.2 km/s

Answer: **(B) 11.2 km/s**

Solution:
Escape velocity: $v_e = \sqrt{(2gR)}$

Where $g = 9.8$ m/s², $R = 6.4 \times 10^6$ m

$v_e = \sqrt{(2 \times 9.8 \times 6.4 \times 10^6)}$

$v_e = \sqrt{(125.44 \times 10^6)}$

$v_e = 11.2 \times 10^3$ m/s = **11.2 km/s**

Q7. In a simple harmonic motion, the acceleration is zero when the displacement is:

(A) Maximum
(B) Zero ✅
(C) Half of maximum
(D) None of these

Answer: **(B) Zero**

Solution:
In SHM, acceleration $a = -\omega^2 x$

When displacement $x = 0$, acceleration $a = -\omega^2(0) =$ **0**

At $x = 0$ (mean position), velocity is maximum and acceleration is zero.
At $x = A$ (extreme position), velocity is zero and acceleration is maximum.

Q8. The refractive index of glass with respect to air is 1.5. The speed of light in glass is:

(A) 3×10^8 m/s
(B) 2×10^8 m/s ✅

(C) 4.5 × 10^8 m/s
(D) 1.5 × 10^8 m/s

Answer: **(B) 2 × 10^8 m/s**

Solution:
Refractive index μ = c/v

Where c = speed of light in vacuum = 3 × 10^8 m/s

v = c/μ = (3 × 10^8)/1.5 = **2 × 10^8 m/s**

Q9. The work done in moving a charge of 5 C across a potential difference of 10 V is:

(A) 2 J
(B) 15 J
(C) 50 J ✅
(D) 0.5 J

Answer: **(C) 50 J**

Solution:
Work done W = qV

W = 5 × 10 = **50 J**

Work done equals the product of charge and potential difference.

Q10. A body of mass 2 kg is moving with a velocity of 3 m/s. The kinetic energy of the body is:

(A) 6 J
(B) 9 J ✅
(C) 12 J
(D) 18 J

Answer: **(B) 9 J**

Solution:
Kinetic energy KE = ½mv^2

KE = ½ × 2 × $(3)^2$

KE = ½ × 2 × 9 = **9 J**

SECTION B: MULTIPLE CORRECT MCQ (Q11 – Q20)

Instructions: Each question has four options. **ONE OR MORE** options may be correct. Full marks are awarded only if all correct options are selected. Marks: +4, No negative marking.

Q11. A projectile is launched at an angle θ with the horizontal. Which of the following statements are correct?

(A) Horizontal component of velocity remains constant ✅
(B) Vertical component of velocity remains constant
(C) The trajectory is parabolic ✅
(D) The acceleration is constant throughout the motion ✅

Answer: **(A), (C), (D)**

Solution:
- **(A) Correct:** No horizontal force acts (ignoring air resistance), so horizontal velocity = $u \cos \theta$ = constant.
- **(B) Incorrect:** Gravity acts vertically, so vertical velocity changes continuously.
- **(C) Correct:** The path of a projectile under uniform gravity is a parabola: $y = x \tan \theta - gx^2/(2u^2\cos^2\theta)$.
- **(D) Correct:** Only gravity acts, so acceleration = g (downward) = constant in magnitude and direction.

Q12. Which of the following are vector quantities?

(A) Displacement ✅
(B) Speed
(C) Torque ✅
(D) Angular momentum ✅

Answer: **(A), (C), (D)**

Solution:
- **Displacement** – has magnitude and direction → **vector** ✅
- **Speed** – has only magnitude → **scalar** ✗
- **Torque** ($\tau = r \times F$) – has magnitude and direction → **vector** ✅
- **Angular momentum** ($L = r \times p$) – has magnitude and direction → **vector** ✅

Q13. For a body undergoing uniform circular motion:

(A) Speed is constant ✅
(B) Velocity is constant
(C) Acceleration is directed towards the center ✅
(D) The net force is directed towards the center ✅

Answer: **(A), (C), (D)**

Solution:
- **(A) Correct:** In uniform circular motion, speed (magnitude of velocity) remains constant.
- **(B) Incorrect:** Direction of velocity changes continuously, so velocity is NOT constant.
- **(C) Correct:** Centripetal acceleration = v^2/r, directed towards the center.
- **(D) Correct:** By Newton's second law, net force = ma, directed towards the center (centripetal force).

Q14. The total energy of a satellite in a stable orbit:

(A) Is negative ✅
(B) Is equal in magnitude to its kinetic energy ✅
(C) Is half of the potential energy in magnitude ✅
(D) Increases with increase in orbital radius

Answer: **(A), (B), (C)**

Solution:
For a satellite in orbit of radius r:
- KE = GMm/(2r)
- PE = −GMm/r
- Total Energy E = KE + PE = −GMm/(2r)

(A) E is negative ✅
(B) |E| = GMm/(2r) = KE ✅
(C) |E| = GMm/(2r) = |PE|/2 ✅
(D) As r increases, |E| decreases (becomes less negative), so total energy increases (becomes less negative). However, the energy approaches zero, meaning the satellite becomes less bound. The total energy does increase algebraically, but **(D)** is technically correct — let me reconsider.

Actually, E = −GMm/(2r). As r increases, E becomes less negative, i.e., E increases. So **(D)** is also correct.

Revised Answer: **(A), (B), (C), (D)**

Correction on (D): E = −GMm/(2r). As r increases, the denominator increases, making the magnitude smaller and E less negative. Since less negative means larger, total energy **does increase** with orbital radius. ✅

Final Answer: **(A), (B), (C), (D)** ✅

Q15. Which of the following statements about the first law of thermodynamics are correct?

(A) It is a statement of conservation of energy ✅
(B) It can be written as ΔQ = ΔU + ΔW ✅
(C) It introduces the concept of entropy
(D) It is applicable to all thermodynamic processes ✅

Answer: **(A), (B), (D)**

Solution:
- **(A) Correct:** The first law is essentially conservation of energy applied to thermodynamic systems.
- **(B) Correct:** Heat supplied = Change in internal energy + Work done by the system.
- **(C) Incorrect:** Entropy is introduced by the **second law** of thermodynamics.
- **(D) Correct:** The first law applies universally to all thermodynamic processes (isothermal, adiabatic, isobaric, isochoric, etc.).

Q16. Which of the following phenomena demonstrate the wave nature of light?

(A) Interference ✅
(B) Diffraction ✅
(C) Photoelectric effect
(D) Polarization ✅

Answer: **(A), (B), (D)**

Solution:
- **(A) Interference** – Superposition of waves → wave nature ✅
- **(B) Diffraction** – Bending of waves around obstacles → wave nature ✅
- **(C) Photoelectric effect** – Demonstrates **particle nature** (photons) ✗
- **(D) Polarization** – Restriction of vibration to one plane → wave nature (transverse waves) ✅

Q17. In a series LCR circuit at resonance:

(A) Impedance is minimum ✅
(B) Current is maximum ✅
(C) Inductive reactance equals capacitive reactance ✅
(D) Power factor is unity ✅

Answer: **(A), (B), (C), (D)**

Solution:
At resonance in series LCR:
- XL = XC, so they cancel out
- Impedance $Z = \sqrt{(R^2 + (XL - XC)^2)} = R$ (minimum)
- Current $I = V/Z = V/R$ (maximum)
- Power factor $\cos \varphi = R/Z = R/R = 1$ (unity)

All four statements are correct. ✅

Q18. Which of the following are correct about electromagnetic waves?

(A) They can travel through vacuum ✅
(B) They are transverse waves ✅
(C) Electric and magnetic fields are perpendicular to each other ✅
(D) They require a material medium for propagation

Answer: **(A), (B), (C)**

Solution:
- **(A) Correct:** EM waves do not require a medium; they travel through vacuum at speed c.
- **(B) Correct:** Both E and B fields oscillate perpendicular to the direction of propagation.
- **(C) Correct:** $E \perp B \perp$ direction of propagation (mutually perpendicular).
- **(D) Incorrect:** EM waves do NOT require a material medium.

Q19. For the photoelectric effect:

(A) The maximum kinetic energy of photoelectrons depends on the frequency of incident light ✅
(B) The number of photoelectrons emitted depends on the intensity of light ✅
(C) There exists a minimum threshold frequency ✅
(D) The maximum kinetic energy depends on the intensity of light

Answer: **(A), (B), (C)**

Solution:

Einstein's photoelectric equation: $KE_{max} = h\nu - \varphi$

- **(A) Correct:** KE_{max} depends on frequency ν.
- **(B) Correct:** More intensity → more photons → more photoelectrons emitted.
- **(C) Correct:** Below threshold frequency ν_0 (where $h\nu_0 = \varphi$), no emission occurs.
- **(D) Incorrect:** KE_{max} depends on frequency, NOT intensity.

Q20. A conducting loop is placed in a time-varying magnetic field. Which of the following are true?

(A) An EMF is induced in the loop ✅
(B) An induced current flows in the loop ✅
(C) The induced current opposes the change in flux ✅
(D) No force acts on the loop

Answer: **(A), (B), (C)**

Solution:
By **Faraday's Law:** $EMF = -d\Phi/dt$. Time-varying B → changing flux → induced EMF ✅

Since the loop is conducting, induced EMF drives an induced current ✅

By **Lenz's Law:** The induced current flows in such a direction as to oppose the change in magnetic flux ✅

(D) Incorrect: The induced current in a magnetic field experiences a force (this is the basis of eddy current braking).

SECTION C: NUMERICAL VALUE (Q21 – Q30)

Instructions: The answer to each question is a **numerical value**. Write the answer up to two decimal places. Marks: +4, No negative marking.

Q21. A ball is dropped from a height of 45 m. How long does it take to reach the ground? ($g = 10\ m/s^2$)

Answer: 3.00 seconds

Solution:
Using $s = ut + \frac{1}{2}gt^2$

$45 = 0 + ½ \times 10 \times t^2$

$45 = 5t^2$

$t^2 = 9$

t = **3.00 s**

Q22. Two resistors of 6 Ω and 3 Ω are connected in parallel. Find the equivalent resistance (in Ω).

Answer: 2.00 Ω

Solution:
For parallel combination:

$1/R_{eq} = 1/R_1 + 1/R_2 = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$

R_{eq} = **2.00 Ω**

Q23. A force of 10 N acts on a body of mass 2 kg initially at rest for 3 seconds. Find the velocity acquired (in m/s).

Answer: 15.00 m/s

Solution:
Acceleration $a = F/m = 10/2 = 5\ m/s^2$

Using $v = u + at = 0 + 5 \times 3$ = **15.00 m/s**

Q24. The focal length of a concave mirror is 15 cm. An object is placed 30 cm from the mirror. Find the image distance (in cm).

Answer: 30.00 cm

Solution:
Using mirror formula: $1/v + 1/u = 1/f$

For concave mirror: $f = -15$ cm, $u = -30$ cm

$1/v + 1/(-30) = 1/(-15)$

$1/v = -1/15 + 1/30 = (-2 + 1)/30 = -1/30$

$v = -30$ cm

Image distance = **30.00 cm** (on the same side as the object, real image)

Q25. A wire of resistance 12 Ω is bent into a circle. The effective resistance between two diametrically opposite points is (in Ω):

Answer: 3.00 Ω

Solution:
When bent into a circle, each semicircular half has resistance = 12/2 = 6 Ω

Between diametrically opposite points, the two halves are in parallel:

$R_{eq} = (6 \times 6)/(6 + 6) = 36/12 =$ **3.00 Ω**

Q26. A car accelerates uniformly from 10 m/s to 30 m/s in 5 seconds. Find the distance traveled (in m).

Answer: 100.00 m

Solution:
Acceleration $a = (v - u)/t = (30 - 10)/5 = 4$ m/s²

Distance $s = ut + \frac{1}{2}at^2 = 10 \times 5 + \frac{1}{2} \times 4 \times 25 = 50 + 50 =$ **100.00 m**

Alternatively: $s = (u + v)/2 \times t = (10 + 30)/2 \times 5 = 20 \times 5 = 100$ m ✅

Q27. The wavelength of a photon with energy 3.1 eV is (in nm): (Use $h = 6.63 \times 10^{-34}$ J·s, $c = 3 \times 10^{8}$ m/s, 1 eV = 1.6×10^{-19} J)

Answer: 400.60 nm

Solution:
$E = hc/\lambda \rightarrow \lambda = hc/E$

$E = 3.1$ eV $= 3.1 \times 1.6 \times 10^{-19} = 4.96 \times 10^{-19}$ J

$\lambda = (6.63 \times 10^{-34} \times 3 \times 10^{8}) / (4.96 \times 10^{-19})$

$\lambda = (19.89 \times 10^{-26}) / (4.96 \times 10^{-19})$

$\lambda = 4.006 \times 10^{-7}$ m = **400.60 nm**

Q28. A capacitor of capacitance 10 μF is charged to 100 V. The energy stored in the capacitor is (in J):

Answer: 0.05 J

Solution:
Energy stored $U = ½CV^2$

$U = ½ \times 10 \times 10^{-6} \times (100)^2$

$U = ½ \times 10^{-5} \times 10^4$

$U = ½ \times 10^{-1}$ = **0.05 J**

Q29. The de Broglie wavelength of an electron accelerated through 150 V is (in Å): (Use standard values)

Answer: 1.00 Å

Solution:
The de Broglie wavelength for an electron accelerated through V volts:

$\lambda = 12.27/\sqrt{V}$ Å

$\lambda = 12.27/\sqrt{150} = 12.27/12.247$ = **1.00 Å**

Q30. A string of length 1 m and mass 0.01 kg is under a tension of 40 N. The speed of the transverse wave on the string is (in m/s):

Answer: 63.25 m/s

Solution:
Linear mass density $\mu = m/L = 0.01/1 = 0.01$ kg/m

Speed $v = \sqrt{(T/\mu)} = \sqrt{(40/0.01)} = \sqrt{4000}$ = **63.25 m/s**

SECTION D: SUBJECTIVE (Q31 – Q40)

Instructions: Answer each question with complete working and explanation. Each question carries 4 marks. In OMR-based evaluation, provide the final numerical answer where applicable.

Q31. State and derive Newton's second law of motion.

Answer & Solution:

Statement: The rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force.

Derivation:
Let a body of mass m be moving with velocity v.

Momentum $p = mv$

According to Newton's second law:

$F \propto dp/dt$

$F = k(dp/dt)$ where k is constant of proportionality

$F = k \times d(mv)/dt$

For constant mass: $F = k \times m(dv/dt) = kma$

In SI system, $k = 1$ (by choice of units)

$\therefore F = ma$

This is the mathematical form of Newton's second law.

Units: Force is measured in Newtons (N). $1\ N = 1\ kg \cdot m/s^2$

Q32. Derive the expression for the time period of a simple pendulum.

Answer & Solution:

Consider a simple pendulum of length L and mass m displaced by a small angle θ.

The restoring force along the arc: $F = -mg \sin \theta$

For small angles ($\theta < 15°$): $\sin \theta \approx \theta = x/L$

$F = -mg(x/L) = -(mg/L)x$

This is of the form $F = -kx$ (SHM), where $k = mg/L$

Angular frequency: $\omega = \sqrt{(k/m)} = \sqrt{(g/L)}$

Time period: $T = 2\pi/\omega$

$T = 2\pi\sqrt{(L/g)}$

Key observations:
- T is independent of mass
- T is independent of amplitude (for small oscillations)
- T depends on length and gravitational acceleration

Q33. A body of mass 5 kg is placed on a rough inclined plane of angle 30°. The coefficient of friction is 0.3. Determine whether the body slides down and find the acceleration if it does. ($g = 10$ m/s²)

Answer & Solution:

Forces on the body:
- Component of gravity along the plane: $mg \sin 30° = 5 \times 10 \times 0.5 = 25$ N
- Normal reaction: $N = mg \cos 30° = 5 \times 10 \times (\sqrt{3}/2) = 43.3$ N
- Maximum static friction: $f = \mu N = 0.3 \times 43.3 = 12.99$ N

Since $mg \sin 30°$ (25 N) > f (12.99 N), the body **slides down**.

Net force $= mg \sin 30° - \mu mg \cos 30°$
$= 25 - 12.99 = 12.01$ N

Acceleration a = Net force / m = 12.01 / 5

$a = 2.40$ m/s²

Q34. Derive the expression for the electric field at a point on the axial line of an electric dipole.

Answer & Solution:

Consider an electric dipole with charges +q and −q separated by distance 2a. Let P be a point on the axial line at distance r from the center.

Electric field due to +q at P:
$E_+ = kq/(r - a)^2$ (away from dipole, towards P)

Electric field due to −q at P:
$E_- = kq/(r + a)^2$ (towards −q, away from P)

Net field: $E = E_+ - E_- = kq[1/(r-a)^2 - 1/(r+a)^2]$

$E = kq[(r+a)^2 - (r-a)^2] / [(r-a)(r+a)]^2$

$E = kq[4ra] / (r^2 - a^2)^2$

$E = k(2qa)(2r) / (r^2 - a^2)^2$

Since dipole moment $p = 2qa$:

$E = 2kpr / (r^2 - a^2)^2$

For $r >> a$:

$E = 2kp/r^3 = (1/4\pi\varepsilon_0)(2p/r^3)$

Direction: Along the dipole moment (from −q to +q).

Q35. Explain the working of a Carnot engine and derive its efficiency.

Answer & Solution:

A Carnot engine operates between a hot reservoir (T_1) and cold reservoir (T_2) through four reversible processes:

1. **Isothermal expansion** (A→B): Gas absorbs heat Q_1 from hot reservoir at T_1
2. **Adiabatic expansion** (B→C): Gas expands without heat exchange, temperature drops to T_2
3. **Isothermal compression** (C→D): Gas rejects heat Q_2 to cold reservoir at T_2
4. **Adiabatic compression** (D→A): Gas is compressed, temperature rises back to T_1

Efficiency:
η = Work done / Heat absorbed = $W/Q_1 = (Q_1 - Q_2)/Q_1$

For a Carnot cycle with ideal gas:
$Q_1/T_1 = Q_2/T_2$

Therefore: $Q_2/Q_1 = T_2/T_1$

$\eta = 1 - T_2/T_1$

Key points:
- $\eta < 1$ always (as $T_2 > 0$ K)
- $\eta = 1$ only if $T_2 = 0$ K (impossible)
- Carnot efficiency is the maximum possible efficiency for any heat engine operating between T_1 and T_2

Q36. Derive the lens maker's formula.

Answer & Solution:

Consider a thin lens of refractive index μ_2 in a medium of refractive index μ_1, with radii of curvature R_1 and R_2.

For refraction at first surface:
$\mu_1/u + \mu_2/v_1 = (\mu_2 - \mu_1)/R_1$ ... (i)

For refraction at second surface (image of first surface acts as object):
$\mu_2/v_1 + \mu_1/v = (\mu_1 - \mu_2)/R_2$ ... (ii)

Adding (i) and (ii):
$\mu_1/u + \mu_1/v = (\mu_2 - \mu_1)[1/R_1 - 1/R_2]$

Dividing by μ_1:
$1/u + 1/v = (\mu_2/\mu_1 - 1)[1/R_1 - 1/R_2]$

Let $\mu = \mu_2/\mu_1$ (relative refractive index)

When $u = \infty$, $v = f$:

$1/f = (\mu - 1)[1/R_1 - 1/R_2]$

This is the **Lens Maker's Formula**.

Q37. A transformer has 500 turns in the primary and 5000 turns in the secondary. If the primary voltage is 220 V, find the secondary voltage, and if the primary current is 5 A, find the secondary current (assuming ideal transformer).

Answer & Solution:

Given: $N_1 = 500$, $N_2 = 5000$, $V_1 = 220$ V, $I_1 = 5$ A

Secondary voltage:
$V_2/V_1 = N_2/N_1$

$V_2 = V_1 \times (N_2/N_1) = 220 \times (5000/500) = 220 \times 10$

$V_2 = 2200$ V

Secondary current:
For ideal transformer: $V_1I_1 = V_2I_2$ (power conservation)

$I_2 = V_1I_1/V_2 = (220 \times 5)/2200$

$I_2 = 0.5$ A

This is a **step-up transformer** (voltage increases, current decreases).

Turns ratio = $N_2/N_1 = 10$

Q38. State and explain Gauss's Law. Use it to find the electric field due to an infinite uniformly charged plane sheet.

Answer & Solution:

Gauss's Law: The total electric flux through any closed surface is equal to $1/\varepsilon_0$ times the total charge enclosed.

** $\oint \vec{E} \cdot d\vec{A} = q_enclosed/\varepsilon_0$**

Application: Infinite charged plane sheet

Let σ = surface charge density of the sheet.

Choose a Gaussian surface: a cylinder (pillbox) with its axis perpendicular to the sheet, extending equally on both sides.

- Area of each flat face = A
- Electric field E is perpendicular to the sheet and points outward on both sides

Flux calculation:
- Through two flat faces: 2EA (E is parallel to dA on both faces)
- Through curved surface: 0 (E is perpendicular to dA)

Total flux: $\Phi = 2EA$

Charge enclosed: $q = \sigma A$

By Gauss's Law: $2EA = \sigma A/\varepsilon_0$

$E = \sigma/(2\varepsilon_0)$

The field is uniform, independent of distance from the sheet, and directed away from the sheet (for positive σ).

Q39. Derive the expression for the magnetic field at the center of a circular current-carrying loop using Biot-Savart's law.

Answer & Solution:

Consider a circular loop of radius R carrying current I. We need to find B at the center O.

Biot-Savart's Law: $dB = (\mu_0/4\pi)(Idl \times \hat{r})/r^2$

For every element dl on the loop:
- $r = R$ (distance from element to center)
- $dl \perp \hat{r}$ (current element is perpendicular to the radius)
- $\sin 90° = 1$

So: $dB = (\mu_0/4\pi)(Idl)/(R^2)$

All dB contributions point in the same direction (perpendicular to the plane of the loop, by right-hand rule).

Total field:
$B = \oint dB = (\mu_0 I)/(4\pi R^2) \oint dl = (\mu_0 I)/(4\pi R^2) \times 2\pi R$

$B = \mu_0 I/(2R)$

Direction: Perpendicular to the plane of the loop (determined by right-hand rule).

Q40. Explain the Bohr model of the hydrogen atom and derive the expression for the radius of the nth orbit.

Answer & Solution:

Bohr's Postulates:
1. Electrons revolve in stationary orbits without radiating energy.
2. Angular momentum is quantized: $L = mvr = nh/(2\pi)$, $n = 1, 2, 3...$
3. Energy is emitted/absorbed when electrons jump between orbits: $\Delta E = h\nu$

Derivation of radius:

For an electron in the nth orbit:

Centripetal force = Electrostatic force:
$mv^2/r = ke^2/r^2$ ... (i)

Quantization condition:
$mvr = nh/(2\pi) \rightarrow v = nh/(2\pi mr)$ ... (ii)

Substituting (ii) in (i):
$m[nh/(2\pi mr)]^2/r = ke^2/r^2$

$m \times n^2h^2/(4\pi^2m^2r^2) \times 1/r = ke^2/r^2$

$n^2h^2/(4\pi^2mr^3) = ke^2/r^2$

$n^2h^2/(4\pi^2m) = ke^2r$

$r = n^2h^2/(4\pi^2mke^2)$

For the nth orbit:
r = n^2a_0

where $a_0 = h^2/(4\pi^2mke^2) = 0.529$ Å (Bohr radius)

For hydrogen: **r = $0.529 \times n^2$ Å**

$r_1 = 0.529$ Å, $r_2 = 2.116$ Å, $r_3 = 4.761$ Å, etc.

SECTION E: TRUE / FALSE (Q41 – Q50)

Instructions: State whether each statement is **TRUE** or **FALSE**. Mark T or F on the OMR sheet. Marks: +4, No negative marking.

Q41. The momentum of a photon is zero since it has no mass.

Answer: FALSE ✗

Solution:
Although a photon has zero rest mass, it possesses momentum.

Photon momentum: $p = h/\lambda = E/c$

This is a consequence of the de Broglie relation. Photons carry momentum, which is demonstrated in phenomena like radiation pressure and Compton scattering.

Q42. In an adiabatic process, no heat is exchanged between the system and surroundings.

Answer: TRUE ✅

Solution:
By definition, an adiabatic process is one in which $\Delta Q = 0$ (no heat transfer).

All changes in energy come from work done: $\Delta U = -\Delta W$

Examples: Rapid compression/expansion, well-insulated systems.

Q43. The resistance of a superconductor is zero below its critical temperature.

Answer: TRUE ✅

Solution:
Superconductivity is a phenomenon where certain materials exhibit exactly zero electrical resistance when cooled below a characteristic critical temperature (Tc).

This was discovered by Heike Kamerlingh Onnes in 1911 with mercury at 4.2 K.

Q44. An object placed between the pole and the focus of a concave mirror always forms a real image.

Answer: FALSE ✗

Solution:
When an object is placed between the pole (P) and focus (F) of a concave mirror, the image formed is:
- **Virtual** (not real)
- **Erect**
- **Magnified**
- Located behind the mirror

This is the principle behind magnifying/shaving mirrors.

Q45. The time period of a geostationary satellite is 24 hours.

Answer: TRUE ✅

Solution:
A geostationary satellite orbits the Earth with a time period equal to Earth's rotation period = 24 hours (approximately 23 hours 56 minutes sidereal).

It orbits in the equatorial plane at an altitude of approximately 35,786 km and appears stationary relative to a point on Earth's surface.

Q46. Kirchhoff's junction rule is based on conservation of charge.

Answer: TRUE ✅

Solution:
Kirchhoff's junction (current) rule states: The algebraic sum of currents at a junction is zero.

$\Sigma I = 0$ (current entering = current leaving)

This is a direct consequence of **conservation of electric charge** — charge cannot accumulate at a junction in a steady-state circuit.

Q47. The wavelength of matter waves increases with increase in velocity of the particle.

Answer: FALSE ✗

Solution:
de Broglie wavelength: $\lambda = h/(mv)$

As velocity v increases, λ **decreases** (inversely proportional).

Therefore, the statement is false. Higher velocity → shorter wavelength.

Q48. In Young's double slit experiment, the fringe width increases when the slit separation is decreased.

Answer: TRUE ✅

Solution:
Fringe width: $\beta = \lambda D/d$

Where:
- λ = wavelength of light
- D = distance from slits to screen
- d = slit separation

$\beta \propto 1/d$

When d decreases, β **increases**. ✅

Q49. The binding energy per nucleon is maximum for iron (Fe-56).

Answer: TRUE ✅

Solution:
The binding energy per nucleon curve peaks at iron-56 (and nearby nuclei like Ni-62).

- For Fe-56: BE/A ≈ 8.8 MeV per nucleon
- Lighter nuclei can undergo **fusion** to approach iron
- Heavier nuclei can undergo **fission** to approach iron

This is why iron is the most stable nucleus.

Q50. A changing electric field produces a magnetic field.

Answer: TRUE ✅

Solution:
This is described by **Maxwell's modification to Ampere's law** (Ampere-Maxwell law):

$\oint \vec{B} \cdot d\vec{l} = \mu_0(I + \varepsilon_0 \, d\Phi E/dt)$

A changing electric field ($d\Phi E/dt \neq 0$) produces a magnetic field, even in the absence of conduction current. This "displacement current" concept was introduced by Maxwell and is essential for the existence of electromagnetic waves.

📊 ANSWER KEY SUMMARY

Section A – Single Correct MCQ

Q	Answer	Q	Answer
1	(B) 20 m	6	(B) 11.2 km/s
2	(A) J·s	7	(B) Zero
3	(A) 50 m	8	(B) 2×10^8 m/s
4	(B) $[MLT^{-2}]$	9	(C) 50 J
5	(B) 1:2	10	(B) 9 J

Section B – Multiple Correct MCQ

Q	Answer
11	A, C, D
12	A, C, D
13	A, C, D
14	A, B, C, D
15	A, B, D
16	A, B, D
17	A, B, C, D
18	A, B, C
19	A, B, C
20	A, B, C

Section C – Numerical Value

Q	Answer	Q	Answer
21	3.00 s	26	100.00 m
22	2.00 Ω	27	400.60 nm
23	15.00 m/s	28	0.05 J
24	30.00 cm	29	1.00 Å
25	3.00 Ω	30	63.25 m/s

Section D – Subjective (Key Answers)

Q	Topic	Key Result
31	Newton's 2nd Law	$F = ma$
32	Simple Pendulum	$T = 2\pi\sqrt{(L/g)}$
33	Inclined Plane	$a = 2.40$ m/s²
34	Electric Dipole	$E = 2kp/r^3$
35	Carnot Engine	$\eta = 1 - T_2/T_1$
36	Lens Maker	$1/f = (\mu-1)(1/R_1 - 1/R_2)$
37	Transformer	$V_2 = 2200$ V, $I_2 = 0.5$ A
38	Gauss's Law	$E = \sigma/(2\varepsilon_0)$
39	Biot-Savart	$B = \mu_0 I/(2R)$

| 40 | Bohr Model | $r_n = 0.529n^2$ Å |

Section E – True/False

Q	Answer	Q	Answer
41	FALSE	46	TRUE
42	TRUE	47	FALSE
43	TRUE	48	TRUE
44	FALSE	49	TRUE
45	TRUE	50	TRUE

Total Marks: 200 | Total Questions: 50 | Duration: 2 Hours

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> **Note:** This document contains all 50 questions with options, correct answers, and detailed solutions. To generate the actual PDF, copy this content into a PDF generator or word processor and export as PDF. The OMR sheet should be generated separately using your OMR evaluation platform.